# ★ Natural Language Processing in Machine Learning.

**Roadmap :**

ML → Supervised → Classification / Regression
ML → Unsupervised → Model

Model → Prediction

| I/P Independent features | | | | Dependent feature O/P |
|---|---|---|---|---|
| $f_1$ | $f_2$ | $f_3$ | $f_4$ | |
| - | - | - | - | - |
| - | - | - | - | - |
| - | - | - | - | - |

Text

Spam classification → Email → Spam / Not Spam

| Email Subject | Email Body | Spam/Ham |
|---|---|---|
| Billionaire | "You won a lottery of Billion $$" | 1 |

⇒ Natural Language Processing

i. Alexa
ii. Google Home

↓ Meaningful vectors

TensorFlow / PyTorch

**Roadmap of NLP.**

BERT
TRANSFORMER
Text Preprocessing → Word Embedding
Neural Nw — RNN, LSTM RNN, GRU RNN → Deep learning
Machine Learning — I/P text → vectors — Text Preprocessing 3 → Word2Vec, Avg Word2Vec
Input text → vectors — Text Preprocessing 2 → BOW, TFIDF, Unigrams, Bigrams
Cleaning the I/P — Text Preprocessing → Tokenization, Lemmatization, Steaming, Stopwords

NLTK, Spacy

Python